

Algebraic geometry and analytic geometry

SGA 1, Exposé XII

English draft

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About this draft

This is an unofficial English translation of Exposé XII, “Géométrie algébrique et géométrie analytique”, from SGA 1 (after unpublished notes of A. Grothendieck). The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ XII

Algebraic geometry and analytic geometry

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Proceeding as in [10], one associates with every scheme X locally of finite type over the field of complex numbers $\mathbf{C}$ an analytic space X^{an} whose underlying set is $X(\mathbf{C})$.

In nos. 2 and 3 of this exposé, we give a “dictionary” between the usual properties of X and of X^{an} and between the properties of a morphism $f: X \rightarrow Y$ and of the associated morphism $f^{\text{an}}: X^{\text{an}} \rightarrow Y^{\text{an}}$.

We then show that the comparison theorems between coherent sheaves on X and X^{an} , established in [10, no. 12] for a projective variety, are still valid when X is a proper scheme.

Finally we prove in no. 5 the equivalence of the category of finite étale coverings of X and of the category of finite étale coverings of X^{an} . As a bonus for the reader, we give a new proof of the theorem of Grauert-Remmert [6], using the resolution of singularities [8].

1. Analytic space associated with a scheme

Let X be a scheme locally of finite type over $\mathbf{C}$. Let Φ be the functor from the category of analytic spaces [4, no. 9] to the category of sets which associates with an analytic space $\mathcal{X}$ the set of morphisms of spaces ringed in $\mathbf{C}$ -algebras $\text{Hom}_{\mathbf{C}}(\mathcal{X}, X)$. We have the following theorem:

THEOREM AND DEFINITION 1.1. *The functor Φ is representable by an analytic space X^{an} and a morphism $\varphi: X^{\text{an}} \rightarrow X$. One says that X^{an} is the analytic space associated with X .*

If $|X^{\text{an}}|$ is the set underlying X^{an} , φ induces a bijection of $|X^{\text{an}}|$ onto the set $X(\mathbf{C})$ of points of X with values in $\mathbf{C}$. Moreover, for each point x of X^{an} , the morphism

$$\varphi_x: \mathcal{O}_{X, \varphi(x)} \longrightarrow \mathcal{O}_{X^{\text{an}}, x},$$

which is necessarily local, gives, by passing to the completions, an isomorphism

$$\widehat{\varphi}_x: \widehat{\mathcal{O}}_{X, \varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\text{an}}, x},$$

In particular the morphism φ is flat.

Let us note that the fact that φ induces a bijection of X^{an} onto $X(\mathbf{C})$ follows from the universal property of X^{an} . On the other hand, we have the following assertions:

- a) If the theorem is true for a scheme Y , the same holds for every subscheme X of Y . Let us first suppose that X is an open subscheme of Y ; if $\psi: Y^{\text{an}} \rightarrow Y$ is the canonical morphism, $\psi^{-1}(X)$ is an open subset of Y^{an} which one endows with the analytic space structure induced by that of Y^{an} . Since every morphism from an analytic space $\mathcal{X}$ into X factors through Y^{an} by the universal property of the latter, hence through X^{an} which is the fiber product $Y^{\text{an}} \times_Y X$, X^{an} is the analytic space associated with X . Finally the assertion concerning the φ_x is evident.

It now suffices to consider the case where X is a closed subscheme of Y . Let I be the coherent $\mathcal{O}_Y$ -Ideal defining X ; then $I \cdot \mathcal{O}_{Y^{\text{an}}}$ is a coherent sheaf of ideals on $\mathcal{O}_{Y^{\text{an}}}$ which defines a closed analytic subspace X^{an} of Y^{an} ; one sees as in the case of an

¹After unpublished notes of A. Grothendieck.

open subscheme that X^{an} is the analytic space associated with X . Let $\varphi: X^{\text{an}} \rightarrow X$ be the canonical morphism. For every point x of X^{an} , the morphism φ_x is none other than the morphism

$$\mathcal{O}_{Y,\psi(x)}/I_{\psi(x)} \longrightarrow \mathcal{O}_{Y^{\text{an}},x}/I_{\psi(x)} \cdot \mathcal{O}_{Y^{\text{an}},x}$$

induced by ψ_x ; its completion

$$\widehat{\varphi}_x: \widehat{\mathcal{O}}_{Y,\psi(x)}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y,\psi(x)} \longrightarrow \widehat{\mathcal{O}}_{Y^{\text{an}},x}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y^{\text{an}},x}$$

is an isomorphism since $\widehat{\psi}_x$ is one, which proves a).

- b) If one has two $\mathbf{C}$ -schemes X_1, X_2 , such that X_1^{an} and X_2^{an} exist, then the same holds for $(X_1 \times X_2)^{\text{an}}$. Indeed, let $\varphi_1: X_1^{\text{an}} \rightarrow X_1$, $\varphi_2: X_2^{\text{an}} \rightarrow X_2$ be the canonical morphisms, and p_1, p_2 the two projections of $X_1^{\text{an}} \times X_2^{\text{an}}$. One deduces formally from EGA I 1.8.1 that $X_1 \times X_2$ is the product of X_1 and X_2 in the category of locally ringed spaces; it follows that the morphisms $\varphi_1 \cdot p_1$ and $\varphi_2 \cdot p_2$ define a morphism $\varphi: X_1^{\text{an}} \times X_2^{\text{an}} \rightarrow X_1 \times X_2$ and that the pair $(X_1^{\text{an}} \times X_2^{\text{an}}, \varphi)$ represents the functor $\mathcal{X} \mapsto \text{Hom}_{\mathbf{C}}(\mathcal{X}, X_1 \times X_2)$.
- c) If $\mathcal{E}^1$ denotes the affine space of dimension 1, i.e. the topological space $\mathbf{C}$ endowed with the sheaf of holomorphic functions, the functor $\mathcal{X} \mapsto \text{Hom}_{\mathbf{C}}(\mathcal{X}, E_{\mathbf{C}}^1)$ is representable by $\mathcal{E}^1$, the canonical morphism $\varphi: \mathcal{E}^1 \rightarrow E_{\mathbf{C}}^1$ being the evident morphism. Indeed, giving a morphism from an analytic space $\mathcal{X}$ into $E_{\mathbf{C}}^1$ is equivalent to giving an element of $\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$, which also amounts to giving a morphism from $\mathcal{X}$ into $\mathcal{E}^1$. One evidently has a bijection $|\mathcal{E}^1| \simeq E^1(\mathbf{C})$, and, for each point $x \in \mathcal{E}^1$, the morphism $\widehat{\varphi}_x$ is none other than the identity morphism of a ring of formal power series in one variable over $\mathbf{C}$.

One deduces from b) and c) that the theorem is true for the affine space $E_{\mathbf{C}}^n$, $n \geq 0$. Using a), one sees that the same holds for every affine scheme X locally of finite type over $\mathbf{C}$. If one no longer supposes X affine and if (X_i) is a covering of X by affine open subsets, it follows from the universal property and from a) that the X_i^{an} glue together and thus define the analytic space X^{an} associated with X .

1.2. Let $f: X \rightarrow Y$ be a morphism of $\mathbf{C}$ -schemes locally of finite type. If $\varphi: X^{\text{an}} \rightarrow X$ and $\psi: Y^{\text{an}} \rightarrow Y$ are the canonical morphisms, it follows from the universal property of Y^{an} that there exists a unique morphism $f^{\text{an}}: X^{\text{an}} \rightarrow Y^{\text{an}}$ such that the diagram

$$\begin{array}{ccc} X^{\text{an}} & \longrightarrow & X \\ f^{\text{an}} \downarrow & & \downarrow f \\ Y^{\text{an}} & \longrightarrow & Y \end{array}$$

is commutative. We have thus defined a functor Φ from the category of $\mathbf{C}$ -schemes locally of finite type to the category of analytic spaces.

The functor Φ commutes with finite projective limits. Indeed, it suffices to see that Φ commutes with fiber products. Now, if X, Y, Z , are schemes locally of finite type over $\mathbf{C}$, it follows from the fact that $X \times_Z Y$ is the fiber product of X and Y over Z in the category of locally ringed spaces that $X^{\text{an}} \times_{Z^{\text{an}}} Y^{\text{an}}$ satisfies the universal property which characterizes $(X \times_Z Y)^{\text{an}}$.

1.3. Let X be a $\mathbf{C}$ -scheme locally of finite type, X^{an} the associated analytic space, $\varphi: X^{\text{an}} \rightarrow X$ the canonical morphism. If F is an $\mathcal{O}_X$ -Module, the inverse image $\varphi^*F = F^{\text{an}}$ is a sheaf of modules on $\mathcal{O}_{X^{\text{an}}}$. One thus defines a functor from the category of $\mathcal{O}_X$ -Modules to the category of Modules on X^{an} . This functor commutes with inductive limits (EGA 0 4.3.2). The sheaf $\mathcal{O}_{X^{\text{an}}}$ being coherent [4, no. 18 §2 th. 2], it transforms coherent sheaves into coherent sheaves (EGA 0 5.3.11). We have moreover:

*[Added in 2003: $E_{\mathbf{C}}^1$ denotes the (algebraic) affine line over $\mathbf{C}$.]

PROPOSITION 1.3.1. *The functor which associates with an $\mathcal{O}_{X^{\text{an}}}$ -Module F its inverse image F^{an} on X^{an} is exact, faithful, conservative.*

Exactness follows from the fact that the morphism $\varphi: X^{\text{an}} \rightarrow X$ is flat (1.1). Let us prove that the functor $F \mapsto F^{\text{an}}$ is faithful. Taking exactness into account, it suffices to show that, if F^{an} is zero, then so is F . Now, for every point x of X^{an} , one then has $F_{\varphi(x)} \otimes_{\mathcal{O}_{X, \varphi(x)}} \mathcal{O}_{X^{\text{an}}, x} = 0$. The morphism $\mathcal{O}_{X, \varphi(x)} \rightarrow \mathcal{O}_{X^{\text{an}}, x}$ being faithfully flat, one has $F_{\varphi(x)} = 0$ for every closed point $\varphi(x)$ of X , and, since X is Jacobson (EGA IV 10.4.8), this implies that F is zero.

The fact that the functor $F \mapsto F^{\text{an}}$ is conservative is formal from exactness and faithfulness.

2. Comparison of the properties of a scheme and of the associated analytic space

PROPOSITION 2.1. *Let X be a $\mathbf{C}$ -scheme locally of finite type, X^{an} the associated analytic space, n an integer. Consider the property P of being*

- (i) *nonempty*
- (i') *discrete*
- (ii) *Cohen-Macaulay*
- (iii) (S_n)
- (iv) *regular*
- (v) (R_n)
- (vi) *normal*
- (vii) *reduced*
- (viii) *of dimension n .*

Then, for X to have the property P , it is necessary and sufficient that X^{an} have it.

Let $\varphi: X^{\text{an}} \rightarrow X$ be the canonical morphism. (i) follows from the fact that one has $|X^{\text{an}}| = X(\mathbf{C})$ (1.1) and from the fact that X is Jacobson (EGA IV 10.4.8). To say that X (resp. X^{an}) is discrete is equivalent to saying that one has $\dim X = 0$ (resp. $\dim X^{\text{an}} = 0$ by [4, no. 19 §4 cor. 6]); (i') therefore follows from (viii).

Let P be one of the properties (ii) to (vii). For X to have the property P , it is necessary and sufficient that P be satisfied at each closed point of X ; indeed, X being excellent (EGA IV 7.8.6 (iii)), the set of points where X satisfies P is open (loc. cit.) and, if this open set contains all the closed points, it is equal to the whole of X . To say that X (resp. X^{an}) has the property P is therefore equivalent to saying that, for every point x of X^{an} , the local ring $\mathcal{O}_{X, \varphi(x)}$ (resp. $\mathcal{O}_{X^{\text{an}}, x}$) has the property P . Since the fact that an excellent local ring has the property P can be seen after passing to the completion, the proposition follows from the isomorphisms $\widehat{\mathcal{O}}_{X, \varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\text{an}}, x}$ in cases (ii) to (vii). The same holds in case (viii), taking into account the relations

$$\dim X = \sup_x \dim \mathcal{O}_{X, \varphi(x)} \quad \dim X^{\text{an}} = \sup_x \dim \mathcal{O}_{X^{\text{an}}, x}$$

where $x \in X^{\text{an}}$. This completes the proof.

PROPOSITION 2.2. *Let X be a $\mathbf{C}$ -scheme locally of finite type, $\varphi: X^{\text{an}} \rightarrow X$ the canonical morphism, T a locally constructible subset of X . Then one has the relation*

$$\varphi^{-1}(\overline{T}) = \overline{\varphi^{-1}(T)}.$$

One may suppose that T is a dense open subset of X . Let H be the reduced closed subscheme of X with underlying space $X - T$; the associated space H^{an} is a closed analytic subspace of X^{an} with underlying space $X^{\text{an}} - \varphi^{-1}(T)$. One must show that every point x of H^{an} belongs to $\overline{\varphi^{-1}(T)}$. Now, at such a point x , the germ of analytic space (X^{an}, x) contains the subgerm (H^{an}, x) , and the latter is defined by a non-nilpotent Ideal of $\mathcal{O}_{X^{\text{an}}, x}$. It then follows from the Nullstellensatz [4, no. 19 §4 cor. 3] that every open neighborhood of x contains points of X^{an} which do not belong to H^{an} , which indeed proves that one has $x \in \overline{\varphi^{-1}(T)}$.

COROLLARY 2.3. *Let X be a $\mathbf{C}$ -scheme locally of finite type, $\varphi: X^{\text{an}} \rightarrow X$ the canonical morphism, T a locally constructible subset of X . For T to be an open subset (resp. a closed subset, resp. a dense subset), it is necessary and sufficient that $\varphi^{-1}(T)$ be so.*

The corollary follows from 2.2 and from the fact that, X being a Jacobson scheme (EGA IV 10.4.8), two locally constructible subsets of X which have the same trace on the very dense set $X(\mathbf{C})$ are equal.

PROPOSITION 2.4. *Let X be a $\mathbf{C}$ -scheme locally of finite type. For X to be connected (resp. irreducible), it is necessary and sufficient that X^{an} be so.*

Suppose X^{an} connected (resp. irreducible). The image $X(\mathbf{C})$ of X^{an} in X is then connected (resp. irreducible). It follows that X is connected (resp. irreducible) since the closed subsets of X and of $X(\mathbf{C})$ correspond bijectively (EGA IV 10.1.2).

Conversely, suppose X connected (resp. irreducible), and let us show that the same holds for X^{an} . One may restrict oneself to the case where X is irreducible. Indeed, suppose X connected. Given a point x of X , the set of points $y \in X$ such that there exists a finite sequence of irreducible closed subschemes $X_1, \dots, X_n$ of X , with $x \in X_1$, $y \in X_n$, $X_i \cap X_{i+1} \neq \emptyset$ for $1 \leq i \leq n-1$, is a set which is both open and closed, hence equal to the whole of X . For a sequence $X_1, \dots, X_n$ as above, one also has $X_i^{\text{an}} \cap X_{i+1}^{\text{an}} \neq \emptyset$ for $1 \leq i \leq n-1$; if one supposes it proved that the X_i^{an} are connected, then the same holds for X^{an} .

From now on we suppose X irreducible. One may moreover suppose X affine. Indeed, if $(U_i)_{i \in I}$ is a covering of X by affine open subsets, any two of these open sets have a nonempty intersection, and the same property is therefore true for the covering $(U_i^{\text{an}})_{i \in I}$ of X^{an} ; if one supposes it proved that the U_i^{an} are irreducible, then the same holds for X^{an} .

One may moreover suppose that X is normal. Indeed, let $\tilde{X}$ be the normalization of X ; since the morphism $\tilde{X} \rightarrow X$ is surjective, so is $\tilde{X}^{\text{an}} \rightarrow X^{\text{an}}$, which proves that, if $\tilde{X}^{\text{an}}$ is irreducible, so is X^{an} .

From now on we suppose X affine and normal. Since the local rings of X^{an} are integral, it amounts to the same to say that X^{an} is irreducible or that it is connected. Indeed, if $\mathcal{F}$ is a closed analytic subset of X^{an} , the set of points x of X^{an} where one has $\text{codim}_x(\mathcal{F}, X^{\text{an}}) = 0$ is a closed analytic subset of X^{an} [4, no. 20 A Cor. 1] which is also open; if X^{an} is connected, this proves that, if one has $\mathcal{F} \neq X^{\text{an}}$, $\mathcal{F}$ is nowhere dense, hence that X^{an} is irreducible. One is thus reduced to showing that X^{an} is connected.

Let

$$i: X \longrightarrow P$$

be a compactification of X , where P is a normal projective $\mathbf{C}$ -scheme and i a dominant open immersion. It then follows from [10, no. 12 th. 1] that P^{an} is connected. Since X^{an} is obtained by removing from P^{an} a nowhere dense closed analytic subset, it follows from 2.5 below that X^{an} is also connected.

LEMMA 2.5. *Let $\mathcal{P}$ be a connected normal analytic space, $\mathcal{Y}$ a nowhere dense closed analytic subset; then $\mathcal{X} = \mathcal{P} - \mathcal{Y}$ is connected.*

When $\mathcal{Y}$ is of codimension ≥ 2 , the proposition follows from [11, no. 3 prop. 4]. In the general case one may suppose, at the cost of removing from $\mathcal{P}$ a closed analytic subset of codimension ≥ 2 , that $\mathcal{P}$ and $\mathcal{Y}$ (considered as a reduced analytic subspace of $\mathcal{P}$) are regular. By the implicit function theorem, every point y of $\mathcal{Y}$ has a neighborhood $\mathcal{U}$ isomorphic to a ball of an affine space $\mathcal{E}^n$, such that $\mathcal{U} \cap \mathcal{Y}$ is defined by the vanishing of a certain number of coordinate functions. This proves that $\mathcal{U} - \mathcal{U} \cap \mathcal{Y}$ is connected, and therefore so is $\mathcal{X}$.

COROLLARY 2.6. *Let X be a $\mathbf{C}$ -scheme locally of finite type; the morphism*

$$\pi_0(X^{\text{an}}) \longrightarrow \pi_0(X)$$

induced by the canonical morphism $X^{\text{an}} \rightarrow X$ is bijective.

3. Comparison of the properties of morphisms

PROPOSITION 3.1. *Let $f: X \rightarrow Y$ be a morphism of $\mathbf{C}$ -schemes locally of finite type, $f^{\text{an}}: X^{\text{an}} \rightarrow Y^{\text{an}}$ the morphism deduced from f on the associated analytic spaces. Let P be the property of being*

- (i) *flat*
- (ii) *net (i.e. unramified)*
- (iii) *étale*
- (iv) *smooth*
- (v) *normal*
- (vi) *reduced*
- (vii) *injective*
- (viii) *separated*
- (ix) *an isomorphism*
- (x) *a monomorphism*
- (xi) *an open immersion.*

Then, for f to have the property P , it is necessary and sufficient that f^{an} have it.

Let us denote by $\varphi: X^{\text{an}} \rightarrow X$ and $\psi: Y^{\text{an}} \rightarrow Y$ the canonical morphisms. Let x be a point of X^{an} , $y = f^{\text{an}}(x)$. The morphisms $\mathcal{O}_{Y^{\text{an}},y} \rightarrow \mathcal{O}_{X^{\text{an}},x}$ and $\mathcal{O}_{Y,\psi(y)} \rightarrow \mathcal{O}_{X,\varphi(x)}$ deduced from f and f^{an} give the same morphism on passing to the completions (1.1). By [2, ch. 3 §5 prop. 4] (resp. EGA IV 17.4.4) it therefore amounts to the same to say that f^{an} satisfies property (i) (resp. (ii)) or to say that f satisfies (i) (resp. (ii)) at each closed point of X . As the set of points of X where (i) (resp. (ii)) is satisfied is an open set (EGA IV 11.1.1 and I 3.3), this proves (i) and (ii), hence also (iii).

Let P be the property (iv) (resp. (v), resp. (vi)). Taking into account 2.1 ((v), (vi), (vii)), it amounts to the same to say that the geometric fibers of f^{an} at the various points y of Y^{an} are regular (resp. normal, resp. reduced) or that the same holds for the geometric fibers of f at the various closed points $\psi(y)$ of Y . The cases (iv) (resp. (v), resp. (vi)) then follow from (i) and from the fact that the set of points of Y where the geometric fibers of f are regular is an open set (EGA IV 12.1.7).

(vii). If f is injective, so is f^{an} . Conversely, suppose f^{an} injective and let us show that the same holds for f . We may assume f of finite type. The morphism f^{an} being injective, the fibers of f at the closed points of Y are radicial; as the set of points of Y whose fiber is radicial is locally constructible (EGA IV 9.6.1) and as Y is a Jacobson scheme, f has all its fibers radicial, hence is injective.

(viii). Let $\Delta: X \rightarrow X \times_Y X$ and $\Delta^{\text{an}}: X^{\text{an}} \rightarrow X^{\text{an}} \times_{Y^{\text{an}}} X^{\text{an}}$ be the diagonal immersions, $\Theta: X^{\text{an}} \times_{Y^{\text{an}}} X^{\text{an}} \rightarrow X \times_Y X$ the canonical morphism. By virtue of 2.3 it amounts to the same to say that $\Delta(X)$ is closed in $X \times_Y X$ or that $\Delta^{\text{an}}(X^{\text{an}})$ is closed in $X^{\text{an}} \times_{Y^{\text{an}}} X^{\text{an}}$.

As an open immersion is nothing other than an injective étale morphism (EGA IV 17.9.1 and [4, no. 13 §1]), (xi) follows from (iii) and (vii). An isomorphism being the same thing as a surjective open immersion, (ix) follows from (xi) and from (3.2 (i)) below. To say that f is a monomorphism is equivalent to saying that the diagonal morphism $\Delta: X \rightarrow X \times_Y X$ is an isomorphism, hence (x) follows from (ix).

PROPOSITION 3.2. *Let X and Y be two $\mathbf{C}$ -schemes locally of finite type, $f: X \rightarrow Y$ a morphism of finite type, $f^{\text{an}}: X^{\text{an}} \rightarrow Y^{\text{an}}$ the morphism deduced from f on the associated analytic spaces. Let P be the property of being*

- (i) *surjective*
- (ii) *dominant*
- (iii) *a closed immersion*
- (iv) *an immersion*

- (v) *proper*²
 (vi) *finite*.

Then, for f to have the property P , it is necessary and sufficient that f^{an} have it.

Let $\varphi: X^{\text{an}} \rightarrow X$ and $\psi: Y^{\text{an}} \rightarrow Y$ be the canonical morphisms.

(i). If f is surjective, for every point y of Y^{an} , $f^{-1}(\psi(y))$ is a nonempty closed subset of X ; it therefore contains at least one closed point, which proves that f^{an} is surjective. Conversely, if f^{an} is surjective, $f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4) which contains all the closed points of Y ; we therefore have $f(X) = Y$.

(ii) follows from 2.2.

(iii). If f is a closed immersion, so is f^{an} by 1.1 a). Conversely, if f^{an} is a closed immersion, so is f by 3.1 (x) and 3.2 (v), since this amounts to saying that f is a proper monomorphism (EGA IV 8.11.5).

(iv). It is clear that, if f is an immersion, so is f^{an} . Conversely, suppose that f^{an} is an immersion, and let T be the image of X in Y , $\bar{T}$ the scheme-theoretic closure of f . We have a factorization of f ,

$$X \xrightarrow{i} \bar{T} \xrightarrow{j} Y,$$

where j is a closed immersion, i the canonical morphism, and we deduce from it the following factorization of f^{an}

$$X^{\text{an}} \xrightarrow{i^{\text{an}}} \bar{T}^{\text{an}} \xrightarrow{j^{\text{an}}} Y^{\text{an}}.$$

As $T = f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4), we have, by 2.2, $\bar{T}^{\text{an}} = \overline{f^{\text{an}}(X^{\text{an}})}$. It follows that $i^{\text{an}}(X^{\text{an}})$ is an open subset of $\bar{T}^{\text{an}}$, hence that $i(X)$ is an open subset of $\bar{T}$. Consider the canonical factorization of i

$$X \xrightarrow{i_1} i(X) \xrightarrow{i_2} \bar{T}.$$

The morphism i_1^{an} is a proper monomorphism, hence so is i_1 by 3.2 (v) and 3.1 (x); this proves that i_1 , hence also f , is an immersion.

(v). Suppose that f is proper and let us show that so is f^{an} . Since the fact that f^{an} is proper is local over Y^{an} , we may assume Y affine. By Chow's lemma (EGA II 5.6.1), we can find a projective Y -scheme X' and a projective surjective morphism

$$g: X' \rightarrow X.$$

The morphism $(fg)^{\text{an}} = f^{\text{an}}g^{\text{an}}$ is projective hence proper, g^{an} is surjective, and it follows from [1, ch. 1 § 10] that f^{an} is proper.

Conversely, suppose f^{an} proper and let us show that so is f . By 3.1 (viii) f is separated. It remains to prove that f is universally closed, and it even suffices to show that f is closed; indeed, for every Y -scheme Y' locally of finite type, the morphism

$$f_{(Y')} = h: X \times_Y Y' \rightarrow Y'$$

will also be closed since h^{an} is proper. Let T be a closed subset of X ; $f(T)$ is a locally constructible set, and we have

$$f^{\text{an}}(\varphi^{-1}(T)) = \psi^{-1}(f(T)).$$

As f^{an} is proper, $\psi^{-1}(f(T))$ is a closed subset of Y^{an} , and it therefore follows from 2.2 that we have

$$\psi^{-1}(\overline{f(T)}) = \psi^{-1}(f(T)).$$

This implies that we have $f(T) = \overline{f(T)}$, i.e. that f is closed, hence that f is proper.

(vi). It amounts to the same to say that a morphism is finite or that it is proper with finite fibers (EGA III 4.4.2 and [4, no. 19 § 5]). As the set of points where the fibers of f are

²We shall say that a morphism of analytic spaces is proper if it is so in the sense of [1, ch. 1 § 10 no. 1] and if it is separated.

finite is locally constructible (EGA IV 9.7.9), the fibers of f are finite if and only if the same holds for the fibers of f^{an} ; (vi) therefore follows from (v).

REMARK 3.3. a) let $f: X \rightarrow Y$ be a morphism of $\mathbf{C}$ -schemes locally of finite type. The fact that f^{an} is a local isomorphism does not imply that the same holds for f . Indeed, if f is étale, f^{an} is étale hence is a local isomorphism [4, no. 13 §1], but this is not necessarily the case for f .

b) Statement 3.2 is not true if one does not assume f of finite type. Let us show for example that f^{an} can be a closed immersion without f being one. Indeed, it suffices to take for X the sum of $\mathbf{Z}$ copies of $\text{Spec } \mathbf{C}$, and for Y the affine line, and for f the morphism obtained by sending the points of X to distinct points of Y forming a discrete subset.

4. Cohomological comparison theorems and existence theorems

The object of this no. is to prove anew the results of [3, no. 2 th. 5 and th. 6]; the latter generalize to the case of a proper scheme the theorems established in [10, no. 12] when X is projective, and extend them to the relative case. More general results, concerning relative schemes proper over an analytic space, are proved in [7, ch. VIII no. 3].

Recall that the Čech cohomology used in [10, no. 12] coincides with the usual cohomology in the algebraic case as in the analytic case (EGA III 1.4.1. and [5, II 5.10]).

4.1. Let $f: X \rightarrow Y$ be a morphism of $\mathbf{C}$ -schemes locally of finite type and consider the commutative diagram

$$\begin{array}{ccc} X^{\text{an}} & \xrightarrow{\varphi} & X \\ f^{\text{an}} \downarrow & & \downarrow f \\ Y^{\text{an}} & \xrightarrow{\psi} & Y. \end{array}$$

If F is an $\mathcal{O}_X$ -Module, we have, for every integer $p \geq 0$, morphisms

$$R^p f_* F \xrightarrow{i} R^p f_*(\varphi_* F^{\text{an}}) \xrightarrow{j} R^p(f \cdot \varphi)_* F^{\text{an}} \xrightarrow{k} \psi_*(R^p f_*^{\text{an}} F^{\text{an}}),$$

where i is deduced from the canonical morphism $F \rightarrow \varphi_* F^{\text{an}}$, and j, k are “edge homomorphisms” of Leray spectral sequences. To the composite $k \cdot j \cdot i$ there is associated a canonical morphism

$$(4.1.1) \quad \theta_p: (R^p f_* F)^{\text{an}} \longrightarrow R^p f_*^{\text{an}}(F^{\text{an}})$$

THEOREM 4.2. *Let $f: X \rightarrow Y$ be a proper morphism of $\mathbf{C}$ -schemes locally of finite type, F a coherent $\mathcal{O}_X$ -Module. Then, for every integer $p \geq 0$ the morphism (4.1.1)*

$$\theta_p: (R^p f_* F)^{\text{an}} \longrightarrow R^p f_*^{\text{an}}(F^{\text{an}})$$

is an isomorphism.

1) *Case where f is projective.* The proof is analogous to that of [10, no. 13]. Let us recall it briefly. One reduces to the case where X is a standard projective space $\mathbf{P}_Y^r$ over Y . Let $\mathcal{Y} = Y^{\text{an}}$, $\mathcal{P} = \mathbf{P}_{\mathcal{Y}}^r$; one first proves that we have

$$f_*^{\text{an}} \mathcal{O}_{\mathcal{P}} = \mathcal{O}_{\mathcal{Y}}, \quad R^p f_*^{\text{an}}(\mathcal{O}_{\mathcal{P}}) = 0 \quad \text{for } p > 0$$

To verify the preceding relations, one can indeed reduce to the case where $\mathcal{Y}$ is a ball $\mathcal{B}$ of an affine space $\mathcal{E}^n$. One considers the “standard covering” $\{\mathcal{U}_i\}$ of $\mathcal{P}$ by $r + 1$ open sets isomorphic to $\mathcal{B} \times \mathcal{E}^r$. As these open sets are Stein, we have, for every integer $p \geq 0$, isomorphisms

$$H^p(\{\mathcal{U}_i\}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} H^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}})$$

One can then express the sections of the structure sheaf $\mathcal{O}_{\mathcal{P}}$ over the open sets $\mathcal{U}_i$ and over their intersections in terms of Laurent series; an easy computation proves that we have

$$H^0(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} H^0(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}}), \quad H^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) = 0 \quad \text{for } p > 0$$

The proof is then completed by copying [10, no. 12 lemma 5], the cohomology groups being replaced by the cohomology sheaves.

2) *Case where f is proper.* One uses EGA III 3.1.2 to reduce to the projective case. Let $\mathcal{K}$ be the category of coherent $\mathcal{O}_X$ -Modules such that θ_p is an isomorphism for every $p \geq 0$. It suffices to prove that, for every exact sequence $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ of which two terms are in $\mathcal{K}$, the same holds for the third, that a direct factor of an object of $\mathcal{K}$ is in $\mathcal{K}$, and that, for every point x of X , one can find an object F of $\mathcal{K}$ such that we have $F_x \neq 0$.

The first condition follows by application of the five lemma to the following commutative diagram, whose rows are exact:

$$\begin{array}{ccccccc} \longrightarrow & (R^p f_* F')^{\text{an}} & \longrightarrow & (R^p f_* F)^{\text{an}} & \longrightarrow & (R^p f_* F'')^{\text{an}} & \longrightarrow & (R^{p+1} f_* F')^{\text{an}} & \longrightarrow \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\ \longrightarrow & R^p f_*^{\text{an}} F'^{\text{an}} & \longrightarrow & R^p f_*^{\text{an}} F^{\text{an}} & \longrightarrow & R^p f_*^{\text{an}} F''^{\text{an}} & \longrightarrow & R^{p+1} f_*^{\text{an}} F'^{\text{an}} & \longrightarrow, \end{array}$$

and one verifies the second condition in an analogous way.

To verify the third condition, one can restrict oneself to the case where X is an irreducible scheme with generic point x . We could have assumed Y noetherian from the start. By Chow's lemma (EGA II 5.6.1), one can find a projective Y -scheme X' and a projective surjective morphism $g: X' \rightarrow X$. On the other hand there exists an integer n such that we have $R^p g_*(\mathcal{O}_{X'}(n)) = 0$ for every $p > 0$ and such that the canonical morphism $g^* g_*(\mathcal{O}_{X'}(n)) \rightarrow \mathcal{O}_{X'}(n)$ is surjective (EGA III 2.2.1). If we set $F = g_*(\mathcal{O}_{X'}(n))$, the sheaf F answers the question. Indeed, we have $F_x \neq 0$; moreover, the Leray spectral sequence

$$R^p f_*(R^q g_*(\mathcal{O}_{X'}(n))) \implies R^{p+q}(f \cdot g)_*(\mathcal{O}_{X'}(n))$$

being degenerate, we have an isomorphism

$$R^p f_* F \xrightarrow{\sim} R^p(f \cdot g)_*(\mathcal{O}_{X'}(n))$$

As in the algebraic case, we have a canonical isomorphism

$$R^p f_*^{\text{an}} F^{\text{an}} \xrightarrow{\sim} R^p(f \cdot g)_*^{\text{an}}(\mathcal{O}_{X'}(n)^{\text{an}}),$$

and the diagram

$$\begin{array}{ccc} (R^p f_* F)^{\text{an}} & \xrightarrow{\sim} & (R^p(f \cdot g)_*(\mathcal{O}_{X'}(n)))^{\text{an}} \\ \theta_p \downarrow & & \downarrow \psi_p \\ R^p f_*^{\text{an}} F^{\text{an}} & \xrightarrow{\sim} & R^p(f \cdot g)_*^{\text{an}}(\mathcal{O}_{X'}(n)^{\text{an}}) \end{array}$$

is commutative. By 1) ψ_p is an isomorphism; hence so is θ_p , which completes the proof.

COROLLARY 4.3. *Let X be a proper $\mathbf{C}$ -scheme, F a coherent $\mathcal{O}_X$ -Module. Then, for every integer $p \geq 0$, the canonical morphism*

$$H^p(X, F) \longrightarrow H^p(X^{\text{an}}, F^{\text{an}})$$

is an isomorphism.

THEOREM 4.4. *Let X be a proper $\mathbf{C}$ -scheme. The functor which associates to every coherent $\mathcal{O}_X$ -Module F its inverse image F^{an} on X^{an} is an equivalence of categories.*

1) *The functor is fully faithful.* Indeed, let F and G be two coherent $\mathcal{O}_X$ -Modules. The canonical morphism

$$\text{Hom}_{\mathcal{O}_X}(F, G) \longrightarrow \text{Hom}_{\mathcal{O}_{X^{\text{an}}}}(F^{\text{an}}, G^{\text{an}})$$

is identified with the canonical morphism

$$H^0(X, \mathbf{Hom}_{\mathcal{O}_X}(F, G)) \longrightarrow H^0(X^{\text{an}}, \mathbf{Hom}_{\mathcal{O}_X}(F, G))$$

(EGA 0_I 6.7.6). As $\mathbf{Hom}_{\mathcal{O}_X}(F, G)$ is coherent, it follows from 4.3 that this morphism is bijective.

2) *The functor is essentially surjective.* When X is projective, the assertion follows from [10, no. 12 th. 3]. The general case reduces to the preceding one by using Chow's lemma (EGA II 5.6.1). Indeed, let X' be a projective $\mathbf{C}$ -scheme, $f: X' \rightarrow X$ a projective surjective morphism, U a dense open subset of X such that f induces an isomorphism $f^{-1}(U) \simeq U$. We argue by noetherian induction on X ; we may therefore assume that, for every coherent sheaf $\mathcal{G}$ on X^{an} such that one can find a closed subset Y of X distinct from X , satisfying the relation $Y^{\text{an}} \supset \text{Supp } \mathcal{G}$, there exists a coherent sheaf G on X such that we have an isomorphism $G^{\text{an}} \simeq \mathcal{G}$.

Let $\mathcal{F}$ be a coherent sheaf of modules on $\mathcal{O}_{X^{\text{an}}}$, and $\mathcal{K}$ and $\mathcal{L}$ the coherent sheaves defined by the condition that the sequence

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{F} \longrightarrow f_*^{\text{an}} f^{\text{an}*} \mathcal{F} \longrightarrow \mathcal{L} \longrightarrow 0$$

be exact. As X' is projective, there exists a coherent $\mathcal{O}_{X'}$ -Module F' such that we have $F'^{\text{an}} \simeq f^{\text{an}*} \mathcal{F}$; one then deduces from 4.2 that we have an isomorphism $(f_* F')^{\text{an}} \simeq f_*^{\text{an}} f^{\text{an}*} \mathcal{F}$. As $\mathcal{K}|_{U^{\text{an}}}$ and $\mathcal{L}|_{U^{\text{an}}}$ are zero, there exist coherent $\mathcal{O}_X$ -Modules K and L such that we have isomorphisms $K^{\text{an}} \simeq \mathcal{K}$, $L^{\text{an}} \simeq \mathcal{L}$. By 1) the morphism $f_*^{\text{an}} f^{\text{an}*} \mathcal{F} \rightarrow \mathcal{L}$ comes from a unique morphism $f_* F' \rightarrow L$; let $I = \text{Ker}(f_* F' \rightarrow L)$. The sheaf $\mathcal{F}$ is then an extension of I^{an} by K^{an} , and it suffices to see that this extension comes by inverse image from an extension of I by K . It therefore suffices to prove that the canonical morphism

$$(*) \quad \text{Ext}_{\mathcal{O}_X}^q(I, K)^{\text{an}} \xrightarrow{\sim} \text{Ext}_{\mathcal{O}_{X^{\text{an}}}}^q(I^{\text{an}}, K^{\text{an}}) \quad q \neq 1$$

is bijective. Now we have isomorphisms $\mathbf{Ext}_{\mathcal{O}_X}^q(I, K)^{\text{an}} \xrightarrow{\sim} \mathbf{Ext}_{\mathcal{O}_{X^{\text{an}}}}^q(I^{\text{an}}, K^{\text{an}})$ for every integer $q \geq 0$ (EGA 0_{III} 12.3.5), and a morphism of spectral sequences

$$\begin{array}{ccc} H^p(X, \mathbf{Ext}_{\mathcal{O}_X}^q(I, K)) & \Longrightarrow & \text{Ext}_{\mathcal{O}_X}^{p+q}(I, K) \\ \downarrow & & \downarrow \\ H^p(X^{\text{an}}, \mathbf{Ext}_{\mathcal{O}_{X^{\text{an}}}}^q(I^{\text{an}}, K^{\text{an}})) & \Longrightarrow & \text{Ext}_{\mathcal{O}_{X^{\text{an}}}}^{p+q}(I^{\text{an}}, K^{\text{an}}). \end{array}$$

This morphism is an isomorphism since, by 4.3, it is so on the terms E_2^{pq} , and this proves the bijectivity of (*).

COROLLARY 4.5. *The functor which associates X^{an} to every proper $\mathbf{C}$ -scheme X is fully faithful.*

We must show that, if X and Y are two proper $\mathbf{C}$ -schemes, the canonical map

$$\text{Hom}_{\mathbf{C}}(X, Y) \longrightarrow \text{Hom}(X^{\text{an}}, Y^{\text{an}})$$

is bijective. Now, to give a morphism from X to Y (resp. from X^{an} to Y^{an}) is equivalent to giving its graph, i.e. a closed subscheme Z of $X \times Y$ (resp. a closed analytic subspace $\mathcal{Z}$ of $X^{\text{an}} \times Y^{\text{an}}$), such that the restriction of the first projection $X \times Y \rightarrow X$ to Z (resp. of $X^{\text{an}} \times Y^{\text{an}} \rightarrow X^{\text{an}}$ to $\mathcal{Z}$) is an isomorphism. As the datum of a closed subscheme of $X \times Y$ (resp. of a closed analytic subspace of $X^{\text{an}} \times Y^{\text{an}}$) is equivalent to that of a coherent sheaf of ideals on $\mathcal{O}_{X \times Y}$ (resp. on $\mathcal{O}_{X^{\text{an}} \times Y^{\text{an}}}$), the corollary follows from 4.4.

COROLLARY 4.6. *Let X be a proper $\mathbf{C}$ -scheme. The functor which associates X'^{an} to every finite (resp. finite étale) scheme X' over X is an equivalence of the category of finite (resp. finite étale) schemes over X with the category of finite (resp. finite étale) analytic spaces over X^{an} .*

Indeed, to give a finite morphism $X' \rightarrow X$ (resp. $X'^{\text{an}} \rightarrow X^{\text{an}}$) is equivalent to giving a coherent sheaf of algebras on $\mathcal{O}_X$ (resp. on $\mathcal{O}_{X^{\text{an}}}$) [4, no. 19 §5 th. 2]. The corollary therefore follows from 4.4 in the case without “resp.”, and the “resp.” case is deduced from it taking into account 3.1 (iii).

5. Comparison theorems for étale coverings

5.0. Let us make precise the notion of finite covering of an analytic space. If $\mathcal{X}$ is an analytic space, one says that an analytic space $\mathcal{X}'$ finite over $\mathcal{X}$ is a finite covering of $\mathcal{X}$ if every irreducible component of $\mathcal{X}'$ dominates an irreducible component of $\mathcal{X}$.

THEOREM 5.1 (“Riemann existence theorem”). *Let X be a $\mathbf{C}$ -scheme locally of finite type, X^{an} the analytic space associated with X . The functor Ψ which, to every finite étale covering X' of X , associates X'^{an} is an equivalence from the category of finite étale coverings of X to the category of finite étale coverings of X^{an} .*

1) *The functor Ψ is fully faithful.* Let X' and X'' be two finite étale coverings of X , and let us prove that the canonical map

$$(*) \quad \text{Hom}_X(X', X'') \longrightarrow \text{Hom}_{X^{\text{an}}}(X'^{\text{an}}, X''^{\text{an}})$$

is bijective. We may assume X' connected. To give an X -morphism from X' to X'' is equivalent to giving a connected component X_i of $X' \times_X X''$ such that the morphism $X_i \rightarrow X'$ induced by the first projection is an isomorphism. Since the connected components of $X' \times_X X''$ correspond bijectively to the connected components of $X'^{\text{an}} \times_{X^{\text{an}}} X''^{\text{an}}$ (2.6) and since a morphism $X_i \rightarrow X'$ is an isomorphism if and only if the same holds for $X_i^{\text{an}} \rightarrow X'^{\text{an}}$, this proves the bijectivity of $(*)$.

2) *The functor Ψ is essentially surjective.* Let $\mathcal{X}'$ be a finite étale covering of X^{an} and let us prove that there exists an étale covering X' of X such that we have an isomorphism $X'^{\text{an}} \xrightarrow{\sim} \mathcal{X}'$. Taking 1) into account, the question is local on X , and we may therefore assume X affine.

a) Reduction to the case where X is normal. We may assume X reduced. Indeed, suppose the theorem proved for X_{red} . The functor which, to a finite étale covering X' of X , assigns the finite étale covering $X'^{\text{an}}_{\text{red}}$ of $X^{\text{an}}_{\text{red}}$ is then an equivalence. Since it is obtained by composing Ψ with the functor Θ which, to a finite étale covering of X^{an} , associates its inverse image on $X^{\text{an}}_{\text{red}}$, and since Θ is fully faithful, this shows that Ψ is an equivalence of categories.

We may assume X normal. Indeed, let $\tilde{X}$ be the normalization of X , $p: \tilde{X} \rightarrow X$ the canonical morphism. Since p is finite, p is an effective descent morphism for the category of étale coverings (IX 4.7). The theorem being assumed proved for $\tilde{X}$, if we set $\tilde{\mathcal{X}}' = \mathcal{X}' \times_{X^{\text{an}}} \tilde{X}^{\text{an}}$, there exist an étale covering $\tilde{X}'$ of $\tilde{X}$ and an isomorphism $\tilde{X}'^{\text{an}} \simeq \tilde{\mathcal{X}}'$. It then follows from 1) that the natural descent datum that we have on $\tilde{\mathcal{X}}'$ lifts to a descent datum on $\tilde{X}'$ relative to $\tilde{X} \rightarrow X$; this proves the existence of an étale covering X' of X such that we have an isomorphism $i: X'^{\text{an}} \times_{X^{\text{an}}} \tilde{X}^{\text{an}} \simeq \tilde{\mathcal{X}}'$ whose inverse images under the two projections of $\tilde{X}^{\text{an}} \times_{X^{\text{an}}} \tilde{X}^{\text{an}}$ are the same. By IX 3.2, whose proof is valid in the analytic case, the morphism $\tilde{X}^{\text{an}} \rightarrow X^{\text{an}}$ is a descent morphism for the category of étale coverings, and consequently i comes from an isomorphism $X'^{\text{an}} \simeq \mathcal{X}'$.

b) Reduction to the case where X is regular. Let U be the open set of regular points of X , $i: U \rightarrow X$, $i^{\text{an}}: U^{\text{an}} \rightarrow X^{\text{an}}$ the canonical morphisms; since X is normal, we have $\text{codim}(X - U, X) \geq 2$. Suppose that there exists an étale covering U' of U such that we have $U'^{\text{an}} \simeq \mathcal{X}'|U^{\text{an}}$, and let us show that then U' extends to an étale covering X' of X such that we have $X'^{\text{an}} \simeq \mathcal{X}'$. It suffices to see that U' extends to an étale covering X' of X ; indeed we shall then have an isomorphism $X'^{\text{an}}|U^{\text{an}} \simeq \mathcal{X}'|U^{\text{an}}$; but, if $\mathcal{F}$ and $\mathcal{G}$ are the coherent sheaves of algebras over $\mathcal{O}_{X^{\text{an}}}$ defining $\mathcal{X}'$ and X'^{an} respectively, the fact that X is normal and that we have $\text{codim}(X - U, X) \geq 2$ implies that the canonical morphisms

$$\mathcal{F} \longrightarrow i_*^{\text{an}}(\mathcal{F}|U^{\text{an}}) \quad \mathcal{G} \longrightarrow i_*^{\text{an}}(\mathcal{G}|U^{\text{an}})$$

are isomorphisms [11, no. 3 prop. 4]. It follows that $\mathcal{F}$ and $\mathcal{G}$, hence also X'^{an} and $\mathcal{X}'$, are isomorphic.

Let $\varphi: X^{\text{an}} \rightarrow X$ be the canonical morphism. Since the problem of extending U' to X is local on X , it suffices to prove that, for every point y of $X^{\text{an}} - U^{\text{an}}$, the étale covering

$U'_{\varphi(y)} = U' \times_X \text{Spec } \mathcal{O}_{X, \varphi(y)}$ of $U_{\varphi(y)} = U \times_X \text{Spec } \mathcal{O}_{X, \varphi(y)}$ extends to $\text{Spec } \mathcal{O}_{X, \varphi(y)}$. Let H be the coherent $\mathcal{O}_U$ -Algebra defining U' . The canonical morphism

$$\alpha: (i_* H)^{\text{an}} \longrightarrow i_*^{\text{an}}(H^{\text{an}}) = \mathcal{F}$$

defines a morphism of sheaves of modules on $\text{Spec } \mathcal{O}_{X^{\text{an}}, y}$:

$$\alpha_y: (i_* H)_y^{\text{an}} \longrightarrow \mathcal{F}_y,$$

whose restriction to $U_y = U_{\varphi(y)} \times_{\text{Spec } \mathcal{O}_{X, \varphi(y)}} \text{Spec } \mathcal{O}_{X^{\text{an}}, y}$ is an isomorphism. But this proves that $H|_{U_y}$ is trivial, hence that $U'_{\varphi(y)}$ extends to $\text{Spec } \mathcal{O}_{X, \varphi(y)}$.

c) Case where X is affine regular. Let

$$j: X \longrightarrow P$$

be a compactification of X , where P is a projective $\mathbf{C}$ -scheme and j a dominant open immersion. Thanks to the resolution of singularities theorem [8], one can find a regular scheme R and a projective morphism $r: R \rightarrow P$ such that r induces an isomorphism $r^{-1}(X) \simeq X$ and $r^{-1}(X)$ is the complement in R of a divisor with normal crossings. Let

$$k: X \longrightarrow R$$

be the canonical immersion. We are going to show that there exists a normal finite covering (5.0) $\mathcal{R}'$ of R^{an} which extends the étale covering X'^{an} . By Proposition 5.3 below, such a covering is unique; the problem of extending X'^{an} is therefore local on R^{an} in a neighborhood of $R^{\text{an}} - X^{\text{an}}$. Now each point of $R^{\text{an}} - X^{\text{an}}$ has an open neighborhood $\mathcal{V}$ isomorphic to a ball of an affine space $\mathcal{E}^n$, such that $\mathcal{V} - \mathcal{V} \cap X^{\text{an}}$ is defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, with $0 \leq p \leq n$. The fundamental group of $\mathcal{U} = \mathcal{V} \cap X^{\text{an}}$ is isomorphic to $\mathbf{Z}^p$, and every étale covering of $\mathcal{U}$ is a quotient of a covering of the form

$$\mathcal{U}'' = \mathcal{U}[T_1, \dots, T_p] / (T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p),$$

where the n_i are integers > 0 , by a subgroup H of the Galois group $\mathbf{Z}/n_1\mathbf{Z} \times \dots \times \mathbf{Z}/n_p\mathbf{Z}$ of $\mathcal{U}''$. Now $\mathcal{U}''$ extends to the regular covering

$$\mathcal{V}'' = \mathcal{V}[T_1, \dots, T_p] / (T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p),$$

of $\mathcal{V}$ on which H acts, and the quotient of $\mathcal{V}''$ by H is the desired extension.

The proof is then completed thanks to 4.6. The covering $\mathcal{R}'$ comes from a finite covering R' of R ; the restriction of R' to X is a covering X' of X such that we have $X'^{\text{an}} \simeq \mathcal{X}'$, and by 3.1(iii) X' is an étale covering of X .

COROLLARY 5.2. *Let X be a connected $\mathbf{C}$ -scheme locally of finite type, $\varphi: X^{\text{an}} \rightarrow X$ the canonical morphism, x a point of X^{an} . Let $\pi_1(X^{\text{an}}, x)$ be the fundamental group of the topological space X^{an} at the point x , and $\pi_1(X, \varphi(x))$ the fundamental group of the scheme X at the point $\varphi(x)$ (V 7). Then $\pi_1(X, \varphi(x))$ is canonically isomorphic to the completion of $\pi_1(X^{\text{an}}, x)$ for the topology of subgroups of finite index.*

Indeed, let $\mathcal{C}$ be the category of finite étale coverings of X^{an} , F the functor from $\mathcal{C}$ to **Ens** which, to every finite étale covering $\mathcal{X}'$ of X^{an} , associates the set of points of $\mathcal{X}'$ over x , and let $\hat{\pi}_1(X^{\text{an}}, x)$ be the profinite group associated with $\mathcal{C}$ and F as explained in V 4. Since every finite étale covering of X^{an} is a quotient of the universal covering by a subgroup of finite index, $\hat{\pi}_1(X^{\text{an}}, x)$ is none other than the completion of $\pi_1(X^{\text{an}}, x)$ for the topology of subgroups of finite index. The corollary therefore follows from 5.1 and V 6.10.

PROPOSITION 5.3. *Let $\mathcal{X}$ be a normal analytic space, $\mathcal{Y}$ a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Then the functor which, to every normal finite covering (5.0) $\mathcal{X}'$ of $\mathcal{X}$, associates its restriction to $\mathcal{U}$ is fully faithful.*

Let $\mathcal{X}'$ and $\mathcal{X}''$ be two normal finite coverings of $\mathcal{X}$. We must show that the canonical map

$$\mathrm{Hom}_{\mathcal{X}}(\mathcal{X}', \mathcal{X}'') \longrightarrow \mathrm{Hom}_{\mathcal{U}}(\mathcal{X}'|_{\mathcal{U}}, \mathcal{X}''|_{\mathcal{U}})$$

is bijective. Let u, v be two $\mathcal{X}$ -morphisms from $\mathcal{X}'$ to $\mathcal{X}''$ whose restrictions to $\mathcal{U}$ are the same, and let us prove that $u = v$. The morphisms u and v coincide on the dense open set $\mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, hence on the underlying topological spaces. By [4, no. 19 §4 cor. 5] this proves that we have $u = v$.

Now let u be a $\mathcal{U}$ -morphism from $\mathcal{X}'|_{\mathcal{U}}$ to $\mathcal{X}''|_{\mathcal{U}}$, and let us show that it extends to the whole of $\mathcal{X}'$. We may assume $\mathcal{X}'$ regular. Indeed, $\mathcal{X}'$ being normal, one can find an open set $\mathcal{V}$ of $\mathcal{X}$ whose complement is an analytic subset of codimension ≥ 2 , such that $\mathcal{X}' \times_{\mathcal{X}} \mathcal{V} = \mathcal{V}'$ is regular. Let $\mathcal{V}'' = \mathcal{X}'' \times_{\mathcal{X}} \mathcal{V}$ and suppose the proposition proved for $\mathcal{V}$. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{V}' & \xrightarrow{i'} & & \mathcal{X}' & \\ & \searrow & & \downarrow f' & \\ & & \mathcal{V}'' & \xrightarrow{i''} & \mathcal{X}'' \\ & \swarrow g'' & & \downarrow f'' & \\ \mathcal{V} & \xrightarrow{i} & & \mathcal{X} & \end{array}$$

To u is associated a morphism of $\mathcal{O}_{\mathcal{V}}$ -Algebras $g''_* \mathcal{O}_{\mathcal{V}''} \rightarrow g'_* \mathcal{O}_{\mathcal{V}'}$, from which one deduces a morphism

$$i_* g''_* \mathcal{O}_{\mathcal{V}''} \longrightarrow i_* g'_* \mathcal{O}_{\mathcal{V}'}$$

Taking into account the isomorphisms $i'_* \mathcal{O}_{\mathcal{V}'} \simeq \mathcal{O}_{\mathcal{X}'}$, $i''_* \mathcal{O}_{\mathcal{V}''} \simeq \mathcal{O}_{\mathcal{X}''}$ [11, no. 3 prop. 4], one deduces from it a morphism of $\mathcal{O}_{\mathcal{X}}$ -Algebras

$$f''_* \mathcal{O}_{\mathcal{X}''} \longrightarrow f'_* \mathcal{O}_{\mathcal{X}'},$$

whence the desired morphism $\mathcal{X}' \rightarrow \mathcal{X}''$.

From now on we assume $\mathcal{X}'$ regular. Let $\mathcal{U}' = \mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, $\mathcal{Y}' = \mathcal{X}' - \mathcal{U}'$. We regard $\mathcal{Y}'$ as a reduced analytic subspace of $\mathcal{X}'$; if $\mathcal{Y}'_1$ is the singular closed set of $\mathcal{Y}'$, we have $\dim \mathcal{Y}'_1 < \dim \mathcal{Y}'$ [4, no. 20 D. Th. 3]. One therefore sees by induction on the dimension of $\mathcal{Y}'$ that we may assume $\mathcal{Y}'$ smooth. Since it suffices to extend u to an open neighborhood of each point of $\mathcal{Y}'$, we may assume, by the implicit function theorem, that $\mathcal{X}'$ is a ball of an affine space $\mathcal{E}^n$ and $\mathcal{Y}'$ the closed set defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, with $0 \leq p \leq n$.

To u we associate a section s of $p: \mathcal{X}' \times_{\mathcal{X}} \mathcal{X}'' \rightarrow \mathcal{X}'$ over $\mathcal{U}'$; after restricting $\mathcal{X}'$, we may assume $p_*(\mathcal{O}_{\mathcal{X}' \times_{\mathcal{X}} \mathcal{X}''})$ generated by elements $x_1, \dots, x_q$ of $\Gamma(\mathcal{X}', p_* \mathcal{O}_{\mathcal{X}' \times_{\mathcal{X}} \mathcal{X}''})$; let $u_1, \dots, u_q \in \Gamma(\mathcal{U}', \mathcal{O}_{\mathcal{X}'})$ be the images under s of $x_1|_{\mathcal{U}'}, \dots, x_q|_{\mathcal{U}'}$. To say that s extends to $\mathcal{X}'$ amounts to saying that the $u_1, \dots, u_q$ extend to sections of $\Gamma(\mathcal{X}', \mathcal{O}_{\mathcal{X}'})$. But, since f is finite, each u_i is a Laurent series in $z_1, \dots, z_p$, with coefficients power series in $z_{p+1}, \dots, z_n$, which satisfy relations of integral dependence. It follows that u_i is bounded, hence is a power series in $z_1, \dots, z_n$, and consequently extends to $\mathcal{X}'$.

One may ask whether the functor introduced in 5.3 is an equivalence of categories. An answer to this question is provided by the theorem of GRAUERT-REMMERT [6], of which we give a proof below using resolution of singularities. One could also have used the theorem of GRAUERT-REMMERT to prove 5.1; this is what was done before [8] was available.

THEOREM 5.4 (GRAUERT-REMMERT theorem). *Let $\mathcal{X}$ be a normal analytic space, $\mathcal{Y}$ a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Let $\mathcal{U}'$ be a normal finite covering of $\mathcal{U}$; assume that there exists a nowhere dense closed analytic subset $\mathcal{S}$ of $\mathcal{X}$ such that the restriction of $\mathcal{U}'$ to $\mathcal{U} - \mathcal{U} \cap \mathcal{S}$ is étale. Then there exists a normal finite covering $\mathcal{X}'$ of $\mathcal{X}$ which extends $\mathcal{U}'$, and $\mathcal{X}'$ is unique up to isomorphism.*

Uniqueness follows from 5.3. The problem of extending $\mathcal{U}'$ is therefore local on $\mathcal{X}$. We may assume $\mathcal{U}$ regular and $\mathcal{U}'$ étale over $\mathcal{U}$. Indeed the set of regular points of $\mathcal{U}$ is an open set $\mathcal{V}$ dense in $\mathcal{X}$ whose complement is an analytic subset [4, no. 20 D th. 2], and it suffices to replace $\mathcal{U}$ by the open set $\mathcal{V} - \mathcal{V} \cap \mathcal{S}$.

Let y be a point of $\mathcal{X} - \mathcal{U}$ and let us show that $\mathcal{U}'$ can be extended to a neighborhood of y . After restricting $\mathcal{X}$ to an open neighborhood of y , it follows from the resolution of singularities theorem [8] that one can find a regular analytic space $\mathcal{X}_1$ and a projective morphism $f: \mathcal{X}_1 \rightarrow \mathcal{X}$ inducing by restriction to $\mathcal{U}$ an isomorphism $\mathcal{U}_1 = f^{-1}(\mathcal{U}) \simeq \mathcal{U}$, such that $\mathcal{U}_1$ is the complement in $\mathcal{X}_1$ of a divisor with normal crossings. Let us show that $\mathcal{U}'$ extends to a normal finite covering of $\mathcal{X}_1$. Since the question is local on $\mathcal{X}_1$, we may assume that $\mathcal{X}_1$ is a ball of an affine space $\mathcal{E}^n$ and that $\mathcal{X}_1 - \mathcal{U}_1$ is defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, with $0 \leq p \leq n$. The étale covering $\mathcal{U}'$ of $\mathcal{U}_1$ is a quotient of a covering of the form

$$\mathcal{U}_2 = \mathcal{U}_1[T_1, \dots, T_p] / (T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p)$$

by a subgroup H of the Galois group of $\mathcal{U}_2$. The covering $\mathcal{U}_2$ extends to the covering

$$\mathcal{X}_2 = \mathcal{X}_1[T_1, \dots, T_p] / (T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p)$$

of $\mathcal{X}_1$ on which H acts, and $\mathcal{X}_2/H$ extends $\mathcal{U}'$ to $\mathcal{X}_1$.

Denote by $\mathcal{X}'_1$ the normal finite covering of $\mathcal{X}_1$ which extends $\mathcal{U}'$, and by $\mathcal{F}_1$ the coherent $\mathcal{O}_{\mathcal{X}_1}$ -Algebra defined by $\mathcal{F}_1$. By the finiteness theorem of GRAUERT-REMMERT [4, no. 15 th. 1.1], $f_*\mathcal{F}_1$ is a coherent $\mathcal{O}_{\mathcal{X}}$ -Algebra. There therefore corresponds to it a finite covering $\mathcal{X}'$ of $\mathcal{X}$, which is moreover normal since $\mathcal{X}'_1$ is, and $\mathcal{X}'$ is the desired extension of $\mathcal{U}'$.

REMARK 5.5. In statement 5.4, one cannot drop the hypothesis on the locus of points where the morphism $\mathcal{U}' \rightarrow \mathcal{U}$ is not étale. For example, let $\mathcal{X}$ be the unit disk of the complex plane, $\mathcal{U}$ the complement of the origin in $\mathcal{X}$, $\mathcal{U}' = \mathcal{U}[T]/(T^2 - \sin 1/z)$, where z is the coordinate function on $\mathcal{X}$. Then $\mathcal{U}'$ is a normal finite covering of $\mathcal{U}$ which does not extend to $\mathcal{X}$. Indeed, suppose that $\mathcal{U}'$ extends to a finite covering $\mathcal{X}'$ of $\mathcal{X}$; the locus of points of $\mathcal{X}$ where the morphism $\mathcal{X}' \rightarrow \mathcal{X}$ is not étale is then a closed analytic set which contains all the points z such that $\sin 1/z = 0$, which is absurd.

One can, however, drop the hypothesis on the singular locus of the morphism $\mathcal{U}' \rightarrow \mathcal{U}$ when we have $\text{codim}(\mathcal{X} - \mathcal{U}, \mathcal{X}) \geq 2$. Indeed, we may assume $\mathcal{U}$ regular. The locus of points of $\mathcal{U}$ where $\mathcal{U}' \rightarrow \mathcal{U}$ is not étale is a divisor of $\mathcal{U}$, and it follows from the theorem of REMMERT-STEIN [9, th. 3] that it is the trace on $\mathcal{U}$ of a divisor of $\mathcal{X}$. Now, in this case, if $\mathcal{A}$ is a coherent $\mathcal{O}_{\mathcal{U}}$ -Algebra such that $\mathcal{U}' = \text{Specan}(\mathcal{A})$, and if $i: \mathcal{U} \rightarrow \mathcal{X}$ is the canonical morphism, it suffices to take $\mathcal{X}' = \text{Specan}(i_*\mathcal{A})$; indeed, we know that $i_*\mathcal{A}$ is coherent [11, no. 1 th. 1].

[Remark 5.6 (added in 2003 (MR)). There exist nontrivial groups G of finite presentation which possess no subgroups of finite index distinct from G , for example the group of G. Higman (cf. J-P. Serre, Arbres et amalgames, Astérisque no. 46, prop. 6, chap. I, § 1). Consequently, if such a group is realized as the topological fundamental group of a scheme V over $\mathbf{C}$, say smooth and projective, the topological space V_{top} underlying V is not simply connected, but V is algebraically simply connected. At present, no such V is known. Let us point out, however, that D. Toledo has constructed smooth projective schemes V over $\mathbf{C}$ whose topological fundamental group is not separated for the topology of subgroups of finite index (the natural morphism $\pi_1(V_{\text{top}}) \rightarrow \pi_1(V_{\text{alg}})$ is not injective). [D. Toledo, Projective varieties with non-residually finite fundamental group, Publ. Math. IHES **77** (1993), p. 103–119].

6. Bibliography

- [1] N. BOURBAKI, Topologie Générale, Hermann, Paris (1960).
- [2] N. BOURBAKI, Algèbre Commutative, Hermann, Paris (1961).

- [3] H. CARTAN. Séminaire E.N.S., Paris (1956-57).
- [4] H. CARTAN. Séminaire E.N.S., Paris (1960-61).
- [5] R. GODEMENT, Théorie des Faisceaux, Hermann, Paris (1958).
- [6] H. GRAUERT and R. REMMERT, Komplexe Räume, Math. Ann. **136** (1958), p. 245–318.
- [7] M. HAKIM, Schémas relatifs, thesis, Paris (1967).
- [8] H. HIRONAKA, Resolution of singularities of an algebraic variety over a field of characteristic zero, Ann. of Math. **39** (1964), p. 109–236.
- [9] R. REMMERT and K. STEIN, Ueber die wesentlichen Singularitäten analytischer Mengen, Math. Ann. **126** (1953), p. 263–306.
- [10] J-P. SERRE, Géométrie algébrique et géométrie analytique, Ann. Inst. Fourier (Grenoble) **6** (1956), p. 1–42.
- [11] J-P. SERRE, Prolongement de faisceaux analytiques cohérents, Ann. Inst. Fourier (Grenoble) **16** (1966), p. 363–374.